	LEMBAR KERJA MANDIRI 2 **COMPUTER VISION** T.A. Genap 2025/2026	**Nilai**	**Paraf Dosen**

A. **Identitas**

Topik : Preprocessing Dataset Foto
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Kelas : Informatika B

B. Tujuan

1. Mahasiswa mampu mengeksplorasi dataset dan analisis awal.
2. Mahasiswa mampu mengstandarisasi ukuran gambar.
3. Mahasiswa mampu melakukan Face Detection dan Croping dari Dataset

C. Instruksi Pengerjaan

- **Tahap 1 – Memount Google Drive dan Mengakses dataset**
   a. Akses link google drive dibawah ini

https://drive.google.com/drive/folders/1PHJ7p-5wgw-gAaAAI4qsa7iJycZE4gN7?usp=sharing

   b. Add Shortcut to Drive / Tambahkan Pintasan
   c. Pilih lokasi **My Drive / Drive Saya**
   d. Mount Google Drive di Colab

```
from google.colab import drive
drive.mount('/content/drive')
```

   e. Copy Dataset ke folder Praktikum

```python
import shutil

source = "/content/drive/MyDrive/DatasetArtis"
destination = "/content/drive/MyDrive/Dataset_Praktikum_Artis"

shutil.copytree(source, destination, dirs_exist_ok=True)

print("Dataset berhasil dicopy")
```

f. Analisis Ukuran Gambar

```python
import os
import cv2

dataset_path = "/content/drive/MyDrive/Dataset_Praktikum_Artis"

sizes = []

for root, dirs, files in os.walk(dataset_path):
    for file in files:
        if file.lower().endswith(('.jpg','.png','.jpeg')):
            img = cv2.imread(os.path.join(root, file))
            if img is not None:
                h, w, c = img.shape
                sizes.append((w,h))

print("Jumlah gambar:", len(sizes))
print("Contoh ukuran:", sizes[:10])
```

Sertakan Screenshot hasil sendiri dan berikan penjelasan mengenai apa yang dilakukan

```python
import os
import cv2

dataset_path = "/content/drive/MyDrive/Dataset_Praktikum_Artis"

sizes = []

for root, dirs, files in os.walk(dataset_path):
    for file in files:
        if file.lower().endswith(('.jpg', '.png', '.jpeg')):
            img = cv2.imread(os.path.join(root, file))
            if img is not None:
                h, w, c = img.shape
                sizes.append((w, h))

print("Jumlah gambar:", len(sizes))
print("Contoh ukuran:", sizes[:10])
```

```
Jumlah gambar: 641
Contoh ukuran: [(620, 349), (449, 488), (3328, 4213), (970, 546), (652, 1000), (62
```

Kode ini digunakan untuk menganalisis dataset gambar dengan cara:

1. Menelusuri seluruh folder dataset.
2. Membaca semua file gambar.
3. Mengambil ukuran setiap gambar.
4. Menampilkan jumlah gambar dan contoh ukuran gambar.

- **Tahap 2 – Standarisasi Ukuran**
    a. Lakukan standarisasi gambar 224 x 224 piksel

```python
import os
import cv2

input_folder = "/content/drive/MyDrive/Dataset_Praktikum_Artis"
output_folder = "/content/drive/MyDrive/Dataset_Resized"

target_size = (224,224)

for root, dirs, files in os.walk(input_folder):
    for file in files:
        if file.lower().endswith(('.jpg','.png','.jpeg')):

            img_path = os.path.join(root, file)
            img = cv2.imread(img_path)
            if img is None:
                continue

            img_resized = cv2.resize(img, target_size)

            relative_path = os.path.relpath(root, input_folder)
            save_dir = os.path.join(output_folder, relative_path)
            os.makedirs(save_dir, exist_ok=True)

            cv2.imwrite(os.path.join(save_dir, file), img_resized)

print("Resize selesai")
```

Sertakan Screenshot hasil sendiri dan berikan penjelasan mengenai apa yang dilakukan

```
import os
import cv2

input_folder = "/content/drive/MyDrive/Dataset_Praktikum_Artis"
output_folder = "/content/drive/MyDrive/Dataset_Resized"

target_size = (224, 224)

for root, dirs, files in os.walk(input_folder):
    for file in files:
        if file.lower().endswith(('.jpg', '.png', '.jpeg')):

            img_path = os.path.join(root, file)
            img = cv2.imread(img_path)
            if img is None:
                continue

            img_resized = cv2.resize(img, target_size)

            relative_path = os.path.relpath(root, input_folder)
            save_dir = os.path.join(output_folder, relative_path)
            os.makedirs(save_dir, exist_ok=True)

            cv2.imwrite(os.path.join(save_dir, file), img_resized)

print("Resize selesai")
```

```
... Resize selesai
```

Kode ini melakukan proses preprocessing dataset gambar, yaitu:

1. Membaca seluruh gambar dalam dataset.
2. Mengubah ukuran semua gambar menjadi 224×224 piksel.
3. Menyimpan hasilnya ke folder baru.
4. Mempertahankan struktur folder dataset asli.

- **Tahap 3 – Standarisasi Ukuran**
    a. Face Detection dan Cropping

```python
import cv2
import os

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

input_folder = "/content/drive/MyDrive/Dataset_Resized"
output_folder = "/content/drive/MyDrive/Dataset_Face_Only"

for root, dirs, files in os.walk(input_folder):
    for file in files:
        if file.lower().endswith(('.jpg','.png','.jpeg')):

            img_path = os.path.join(root, file)
            img = cv2.imread(img_path)
            if img is None:
                continue

            gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
            faces = face_cascade.detectMultiScale(
                gray,
                scaleFactor=1.3,
                minNeighbors=5
            )

            for (x,y,w,h) in faces:
                face = img[y:y+h, x:x+w]
                face = cv2.resize(face,(224,224))

                relative_path = os.path.relpath(root, input_folder)
                save_dir = os.path.join(output_folder, relative_path)
                os.makedirs(save_dir, exist_ok=True)

                cv2.imwrite(os.path.join(save_dir,file),face)

print("Face extraction selesai")
```

Sertakan Screenshot hasil sendiri dan berikan penjelasan mengenai apa yang dilakukan

```python
import cv2
import os

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

input_folder = "/content/drive/MyDrive/Dataset_Resized"
output_folder = "/content/drive/MyDrive/Dataset_Face_Only"

for root, dirs, files in os.walk(input_folder):
    for file in files:
        if file.lower().endswith(('.jpg', '.png', '.jpeg')):

            img_path = os.path.join(root, file)
            img = cv2.imread(img_path)
            if img is None:
                continue

            gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
            faces = face_cascade.detectMultiScale(
                gray,
                scaleFactor=1.3,
                minNeighbors=5
            )

            for (x, y, w, h) in faces:
                face = img[y:y+h, x:x+w]
                face = cv2.resize(face, (224, 224))

                relative_path = os.path.relpath(root, input_folder)
                save_dir = os.path.join(output_folder, relative_path)
                os.makedirs(save_dir, exist_ok=True)

                cv2.imwrite(os.path.join(save_dir, file), face)

print("Face extraction selesai")
```

```
Face extraction selesai
```

Kode ini melakukan proses Face Extraction pada dataset gambar dengan langkah:

1. Membaca seluruh gambar dalam dataset.
2. Mendeteksi wajah menggunakan Haar Cascade Classifier.
3. Memotong area wajah dari gambar.
4. Mengubah ukuran wajah menjadi 224×224 piksel.
5. Menyimpan hasilnya ke folder baru.

Hasil dataset ini biasanya digunakan untuk training model pengenalan wajah (Face Recognition) dalam Machine Learning atau Deep Learning.

--- Selamat Mengerjakan ---